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Multimodal cyberbullying detection in Hinglish memes using a classroom framework based on large language models

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Abstract

The widespread use of social media has led to the emergence of harmful memes, which can perpetuate hate speech and cyberbullying. Detecting such harmful content is crucial for maintaining a safe online environment. Cyberbullying detection in code-mixed languages, particularly Hinglish (Hindi-English), poses significant challenges due to the complexity of multimodal content. This paper presents a novel approach for multimodal cyberbullying detection in Hinglish memes, leveraging the power of Large Language Models (LLMs) and a teacher-student learning based classroom framework. The proposed method involves generating explanations from both harmless and harmful perspectives using LLMs, followed by a final decision made by a smaller language model, mT5. This gives our model the ability to use multimodal explanations derived from both harmful and harmless arguments to execute dialectical reasoning over complex and implicit harm-indicative patterns. The model is trained on the MultiBully dataset, a benchmark dataset of Hinglish memes annotated with bully and non-bully labels. Experimental results demonstrate the effectiveness of the proposed approach, achieving an accuracy of 73.31% and an F1 score of 73.23%, outperforming several baselines. The paper concludes by discussing the implications of the findings and potential future directions for research in this area.

Keywords Cyberbullying detection, Hindi English mixed, Deep learning, Transformer, Psychological, LLM knowledge transfer, MT5

1 Introduction

The increasing prevalence of social media platforms has revolutionized the way people communicate and share information, fostering global connectivity and enabling the rapid dissemination of ideas. However, this digital landscape has also become a breeding ground for harmful content, including cyberbullying and hate speech [1]. Among the various forms of online expression, memes, which combine images and text to convey humorous or satirical messages, have gained immense popularity. Unfortunately, this popularity has also been exploited to spread harmful messages, often targeting vulnerable individuals or groups. The detection of such harmful content is crucial to



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maintaining a safe online environment and protecting users from the harmful effects of cyberbullying.

The detection of harmful memes presents a unique challenge due to their multimodal nature [2], where the meaning is conveyed through both visual and textual elements. Traditional text-based approaches for cyberbullying detection often fall short in capturing the nuances of these multimodal messages. Moreover, the implicit and often subtle nature of harmful memes makes their identification even more difficult, as the harmful content may not be explicitly stated but rather implied through the combination of images and text.

Existing research on cyberbullying detection has primarily focused on English language content [3], overlooking the linguistic diversity of the internet. In multilingual societies like India, code-mixing, the practice of blending multiple languages in communication, is prevalent. Hinglish, a blend of Hindi and English, is a prime example of such code-mixing and is commonly used in online interactions. This linguistic diversity poses additional challenges for cyberbullying detection models, as they need to be adapted to handle the unique characteristics of code-mixed language, such as the use of different scripts, grammar, and vocabulary.

The implicit nature of harmful memes, coupled with the linguistic diversity of the internet, necessitates the development of sophisticated approaches that can effectively identify and mitigate the spread of such content and enhance social integration. Using a knowledge transfer framework and the powers of Large Language Models (LLMs), this research presents a unique method for multimodal cyberbullying detection in Hinglish memes. The suggested approach uses LLMs to produce explanations from positive as well as negative viewpoints, with a smaller language model, mT5, making the ultimate judgement.

The use of LLMs in the proposed approach allows for a deeper understanding of the meme's meaning by generating explanations that consider both the visual and textual elements. The teacher-student framework, inspired by human reasoning and argumentation, enables the model to consider multiple perspectives and understand the reasons before making a final decision. This approach not only improves the accuracy of cyberbullying detection but also provides valuable insights into the reasoning behind the model's decisions, enhancing explainability and transparency.

The MultiBully dataset [4], a benchmark dataset of Hinglish memes annotated with bully and non-bully labels, is utilized in this study to evaluate the effectiveness of the proposed approach. This dataset, with its diverse collection of multimodal memes, provides a valuable resource for training and testing cyberbullying detection models in the Hinglish language. By leveraging the rich linguistic and visual information present in this dataset, the proposed approach aims to advance the state-of-the-art in cyberbullying detection and contribute to a safer online environment.

This paper proposes a novel framework for detecting cyberbullying in Hinglish memes, leveraging a teacher-student knowledge transfer mechanism between large language models (LLMs). Inspired by classroom learning [5], we have created this original framework by integrating mT5, a multilingual variant of the T5 model, tailored for handling code-mixed language inputs. Our approach focuses solely on binary classification (bullying vs. non-bullying) and demonstrates significant improvements over existing baselines.

The remainder of this article is organized as follows. Section 2 introduces the related work of Cyberbullying Detection and Code-Mixed Hinglish Classification, Sect. 3 details the methodology and structure of our framework and the experimental procedures we apply to it, Sect. 4 discusses and analyses the experimental results and Sect. 5 demonstrate the main conclusion in summarized form.

2 Related work

This section discusses the effectiveness of statistical, Machine Learning, and Deep Learning based approaches for cyberbully detection. In addition, recent LLM-based approaches used to detect cyberbullying events are also discussed in this section.

2.1 Cyberbullying detection

Cyberbullying detection has been an active area of research, with various approaches [6–8] proposed to address this issue. Early studies primarily focused on text-based methods, utilizing machine learning algorithms like Support Vector Machines (SVM) and Naive Bayes to classify text as bully or non-bully [9]. They applied these techniques to a dataset of 4500 YouTube Comments achieving an overall accuracy of 66.70% with the SVM Classifier and 63% with Naive Bayes classifier.

With the advent of deep learning, models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) [10, 11] have been employed to capture the nuances of language and improve detection accuracy. The introduction of transformer-based models like BERT has further advanced the field, enabling the capture of contextual information and long-range dependencies in text. A BERT-based approach called cyberBERT, or cyberbully identification, was created in 2020 by Paul et al. [12]. They achieved state-of-the-art findings by evaluating cyberBERT on three benchmark datasets: Wikipedia (100k posts), Twitter (16k posts), and Formspring (12k posts) [13].

However, these text-based approaches often fall short when dealing with multimodal content like memes, where the meaning is conveyed through both text and images. Some studies have attempted to incorporate visual features into cyberbullying detection models, but these efforts have been limited in their effectiveness. Traditionally, research has used attention-based mechanisms and multimodal fusion techniques to classify harmful memes using classical two-stream models that integrate text and vision features learned from text and image encoders [6, 8, 14, 15]. Optimising pre-trained multimodal models for the job was the focus of another branch [16–19]. Additionally, recent studies have attempted to investigate the application of harmful target disentanglement[20] (Lee et al., 2021), ensemble approaches [18, 21, 22], and data augmentation strategies[23, 24].

Researchers were further motivated to create methods for identifying harmful memes in hate speech by Facebook's Hateful Memes Challenge[14, 25]. In a more recent study, [7] provided a formal definition of the term "harmful meme" and showed how contextual elements influence its meaning.

Furthermore, most research has concentrated on English language content, neglecting the linguistic diversity of the internet. In multilingual societies, code-mixing is a common phenomenon, and cyberbullying detection models need to be adapted to handle such linguistic variations. Some studies have explored the use of multilingual language models like mBERT to handle code-mixed text, but these models often require

large amounts of training data and may not generalize well to different code-mixing patterns[26].

2.2 Large language models

Recent advancements in Large Language Models (LLMs) have shown promising performance in various natural language processing tasks, including text generation, translation, and question answering[27–31]. These models, trained on massive amounts of text data, have the potential to capture the subtle nuances of language and understand the context of online interactions. However, their application to cyberbullying detection in memes, particularly in code-mixed languages like Hinglish, remains relatively unexplored.

The proposed approach in this paper aims to bridge this gap by leveraging the power of LLMs and a knowledge transfer framework for multimodal cyberbullying detection in Hinglish memes. The teacher-student framework, inspired by human learning, reasoning and argumentation, allows for the generation of explanations from both harmless and harmful perspectives. This approach not only improves the accuracy of cyberbullying detection but also provides valuable insights into the reasoning behind the model's decisions, enhancing explainability and transparency.

In addition to the challenges mentioned above, the detection of cyberbullying in memes is further complicated by the presence of sarcasm and humor. Sarcasm often involves saying the opposite of what is meant, while humor can be used to mask harmful intent. These factors make it difficult for traditional machine learning models to accurately identify cyberbullying in memes. The proposed method in this paper aims to address this issue by leveraging the capabilities of LLMs to understand the context and intent behind the text and images in memes.

3 Proposed methodology

This section demonstrates the problem statement for cyberbullying meme detection at the initial phase. Based on this, the proposed Classroom Framework, including teacher and student modules, is discussed further.

3.1 Problem statement

We consider a cyberbullying meme detection dataset to consist of a series of memes, each described by a pair $M = \{I, T\}$, where I stands for the image and T for the accompanying text. To address this task, we treat it as a language generation problem, building on previous studies to harness the capabilities of pre-trained large language models (LLMs).

By extracting reasoning from LLMs, which provide explanations from both bullying and non-bullying viewpoints, we aim to strengthen the model's capacity to reason effectively. Our approach takes both the image (I) and text (T) as inputs and outputs a label, $u \in \{bully, nonbully\}$, to determine whether the meme conveys bully or non-bully content. This strategy emphasizes dialectical reasoning, where conflicting perspectives are examined to achieve a more nuanced decision in identifying harmful memes. We base this concept on how students learn in a classroom after the teachers discuss both viewpoints in front of the class and allow the students to make their own decisions. The explanatory foundation for the model's conclusion might readily arise from the LLM

reasoning relating to our model's harmfulness forecast. Figure 1 depicts our framework's overview. The Multimodal Teacher module with LLMs and the Multimodal Student module with Small LM make up this set.

3.2 Multimodal teacher module

It is possible to produce natural language explanations that explore the hidden meaning of memes with the help of LLMs, which makes it easier to assess whether or not memes are detrimental. However, it is highly possible that preconceptions may distort the rationales derived directly from LLMs, which could result in incorrect labels and impair detection performance. We develop an approach that encourages a multimodal dialogue between large language models (LLMs) about the content of memes. In this setup, two agents or "teachers" offer distinct reasoning—one from a positive, the other from a negative perspective—prompting the model to investigate diverse angles and reasoning paths. This method is designed to promote a deeper exploration of varying viewpoints on the memes. Subsequently, by signalling which logic is more plausible, a third agent or teacher will be able to deduce meme harmfulness based on this kind of nonpartisan thinking chain. This section focuses on the discussion generation prompting approach.

In order to prompt the vision LLMs, or LLaVA, taking a meme pair $M = \{I, T\}$, we construct a prompt definition p^k comprising of the text T , the meme image I and the probable harmfulness label $k \in \{b, nb\}$, which indicates "bully" or "nonbully" as observable qualities. Every agent will provide a rationalisation r^k that elicits reasoning expertise about how to deduce a certain harmfulness label (k) from the interaction between the meme image (I) and text (T). The prompt template p^k which we use is:

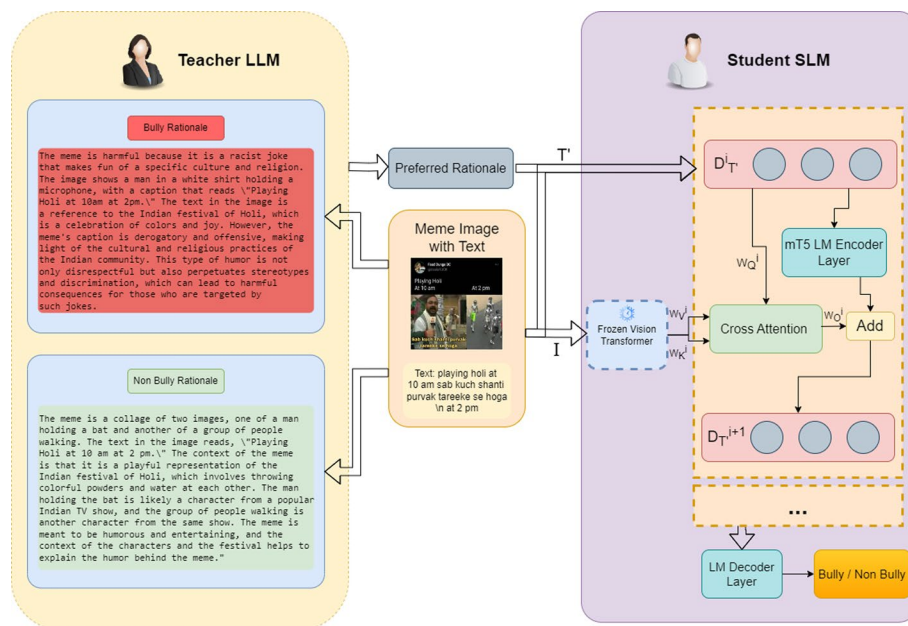


Fig. 1 Pipeline of our proposed Classroom Framework. In this, at first, a multimodal discussion amongst teacher LLMs is initiated in order to produce opposing arguments from the perspectives of the nonbully (green) and bully (red) parties. Next, using multimodal meme inputs, the resulting rationales are utilised to train a task-specific Small Language Model student

"Considering the image "[I]", which incorporates the text "[T]", provide a concise analysis connecting the text and the image. Utilize your knowledge of the surrounding context and general understanding to clearly explain why the image is reasoned as [k]."

Incorporating the potential harmfulness label into each definition can activate detailed contextual information. This allows for a clearer representation and comparison of the subtle contexts tied to various harmfulness categories in memes. By offering a foundation to independently advance the harmfulness label in each discussion, the model is better equipped to analyze and justify the true level of harmfulness through contrasting dialectical perspectives.

3.2.1 Multimodal comparison with LLM

Motivated by dialectical reasoning and Tree of Thought (ToT) prompting approaches,cao2023probabilistic,we take use of the opposing viewpoints in the multimodal discussion between agents to give an LLM with emergent skills dialectical reasoning chains so that it can determine if memes are bully or not. Consequently, we insert the opposing arguments into the LLM's reasoning process in order for it to deduce the projected harmfulness label. To make sure that no more variables, such various inductive biases, are added, we choose to use the same LLM for the third teacher agent, even if any LLM with capabilities equivalent to or greater than the first two agents can be used. Specifically, we used this dialectical prompting approach to turn the rationales r^b and r^nb into a contending thought chain, serving as the LLM Agent's textual prompt:

"Provided with the meme, consisting of the Text: [T] embedded in the Image, and the following two options: (1) Non-Bully: [r^nb]; (2) Bully: [r^b], is this meme bully or non-bully? "

This prompting strategy is critical for distinguishing the value of the Teacher-Student setup from simple external knowledge augmentation. By forcing the LLM to choose between a "Bully" and "Non-Bully" rationale (as shown in equation 1), we do not simply concatenate external knowledge; rather, we filter it. This "preference selection" acts as a gatekeeper, retaining external knowledge only when it aligns logically with the visual and textual evidence, thereby reducing the noise that typically accompanies raw knowledge retrieval.

The LLM agent uses the input prompt to determine which of the two relevant explanations from the argument is more logical. Based on this inference, the harmfulness label is given precedence. This might be utilised as the only output or as an additional source of information for a precise classification model that is discussed in the next section.

3.3 Multimodal student module with mT5

The LLM agent benefits from the wide range of information obtained through multimodal discussions but its inferences are often unreliable due to the inherent constraints of large language models. Additionally, fine-tuning the LLM agent for this specific task is inefficient because of the large number of parameters involved. To tackle this issue, we suggest enhancing meme classification accuracy by fine-tuning a more compact student language model, mT5, specifically designed to differentiate between bullying and non-bullying material. This method leverages prior knowledge in the form of rationales derived from conflicting arguments about harmfulness. By doing so, it aims

to strengthen multimodal interactions, allowing the rationales from the LLM agents to integrate effectively with the diverse multimodal elements within the memes.

In our Meme pair $M = \{I, T\}$, we first join the text of the meme (T) and the resultant rationales which become the input array for our mT5 agent. In a manner akin to the rationale input sequence used by the LLM teacher, we can exploit the bully inference generated by the LLM agent while fine-tuning the mT5 model. In this case, we prioritize the reasoning deemed more reasonable by the LLM agent. To achieve this, we set the input text (T') as:

$$T' = [T, r^{(1)}, r^{(2)}], \quad (1)$$

where $[,]$ denotes the concatenation operation and where $(1) > (2)$, indicating that the LLM teacher prefers $(r^{(1)})$ to $(r^{(2)})$, and $(1), (2) \in b, nb$. By reorganizing the rationale order according to the LLM teacher's previous evaluation, we effectively integrate the knowledge and insights derived from the LLM. This reordering enables the mT5 model to prioritize difficult training instances that were incorrectly classified by the LLM teacher, while minimizing attention on simpler cases that were already accurately identified. We hypothesize that this approach enables the model to refine its understanding of the memes and rectify the LLM's misperceptions using the training data.

Next, we embed the meme image (I) and the input text (T') to obtain their encoding vectors as follows:

$$D_{T'}^0 = \text{TT}(T'), \quad D_I = \text{VT}(I), \quad (2)$$

where $(\text{TT}(\cdot))$ represents the text embedding layer of the student LM Encoder, and $(D_{T'}^0 \in R^{n \times m})$ are the token embeddings produced by the Transformer encoder's embedding layer. Here, (r) is the text length of (T') , and (m) is the size of the hidden states. Leveraging the Relative Position Encoding of LMs, the LLM output can be integrated into the mT5 agent based on the relative position information of the input sequence. $(\text{VT}(\cdot))$ is the Vision Extractor based on a pre-trained CLIP image transformer with frozen parameters, used to extract patch-level features from the image with (t) patches, which are then projected into visual representations $(D_I \in R^{t \times m})$.

In our setup, we use a CLIP variant that expects images resized to a fixed resolution of 224×224 . While this limits the model's ability to resolve fine-grained visual details or read small text embedded directly within the image pixels, this limitation is mitigated by the explicitly provided text. The MultiBully dataset, maity2022multitask explicitly provides all text (T) present in the meme, which serves as a direct input to both the Teacher and Student modules, reducing reliance on the vision encoder for textual understanding.

To improve semantic alignment between the meme sample and the harmfulness rationales, thereby enhancing cross-modal context comprehension, we utilize a cross-attention mechanism. This mechanism aligns the visual representations with the textual ones, enabling the effective fusion of multimodal information within the mT5 model.:

$$D_I^i = \text{softmax} \left(\frac{Q_{T'} K_I^B}{\sqrt{m_k}} \right) V_I, \quad (3)$$

where the query, key, and value are defined as $(Q_{T'}, K_I, V_I = D_{T'}^i W_Q^i, D_I W_K^i, D_I W_V^i)$, and $(W_Q^i, W_K^i, W_V^i \in R^{m \times m_k})$ are trainable weights. $(D_{T'}^i)$ represents the input hidden

states of the (i)-th LM Encoder layer, and (D_I^i) represents the attended visual features. We then fuse (D_I^i) with $(D_{T'}^i)$ to obtain the interplay representations for a meme:

$$(D_{T'}^{i+1}) = LME^i(D_{T'}^i) + D_I^i W_O^i, \quad (4)$$

where $(LME^i(\cdot))$ is the (i)-th layer of the LM Encoder, (W_O^i) denotes the linear projection, and $(0 \leq i \leq L - 1)$ given the total (L) layers in the LM Encoder. We denote $(D' = D_{T'}^L)$ as the final interplay representations.

Model Training. The interplay representations $(D' \in R^{n \times m})$ are fed into the LM Decoder, implemented as a Transformer-based decoder, to generate the predicted label. By adapting the generative objective [49] to pre-trained LMs, the mT5 student agent can leverage prior reasoning knowledge acquired during pre-training to better deduce harmfulness predictions. Specifically, our mT5 student agent, denoted as $f(I, T')$, is trained by minimizing the loss:

$$L = CE(f(I, T'), u), \quad (5)$$

where $(CE(\cdot))$ represents the cross-entropy loss between the generated label token and the ground-truth harmfulness label (u) . Our student module, led by the previous preference of the LLM agent, learns to recognise the different relative positions among the harmfulness rationales. By comparing with the ground-truth labels, this motivates the model to learn. In order to do this, we use Relative Position Encoding to initialise our model using the mT5 encoder-decoder architecture. Our mT5 judge can improve overall detection performance by attending to implicit harm-indicative patterns in the rationales that the LLM agent misinterpreted during fine-tuning. Simultaneously, the rationale corresponding to the final prediction serves as a basis for explaining the decision in natural language.

4 Experiments and results

This section provides a comprehensive overview of the dataset selection process, detailing the criteria and sources used to compile the data. It also describes the experimental setup, including the methodology, tools, and parameters employed to conduct the experiments. Additionally, the evaluation metrics are presented, explaining how the results are measured, validated, and compared to ensure the accuracy of the findings.

4.1 Dataset description

To evaluate the performance of the proposed model, we have experimented with publicly available Multibully data.¹ Fig. 2 shows four memes (M1–M4) sampled from Multibully

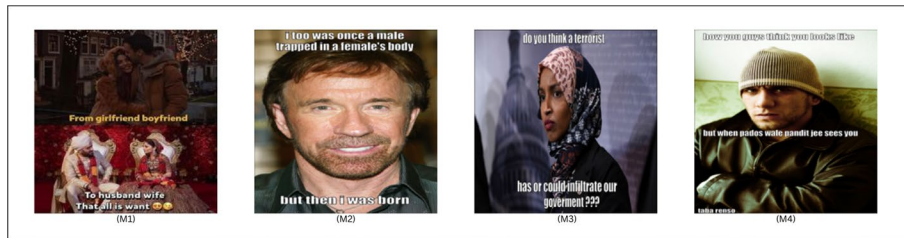


Fig. 2 A few samples from publicly available multimodal meme dataset, Multibully (M1 - M4) [4]

¹ <https://github.com/Jhprince/MultiBully?tab=readme-ov-file>.

dataset[4] illustrates how detecting harmful content in memes requires a combined understanding of both textual and visual elements.

In the first meme (M1), the text expresses a progression from casual dating to marriage, accompanied by images of couples in different relational contexts. While seemingly harmless, such content could be used to pressure individuals or ridicule their personal choices depending on the context, highlighting how subtle insinuations require a nuanced analysis that considers both text and imagery. The second meme (M2) makes light of transgender experiences by mocking gender identity, using the face of a well-known male figure to intensify the ridicule. Here, the textual disparagement combined with the visual cue reinforces harmful stereotypes, clearly constituting cyberbullying. The third meme (M3) pairs a question insinuating terrorism with the image of a woman wearing a hijab, thereby promoting religious prejudice and fear. The multimodal alignment between the text's insinuation and the visual representation of a specific community is what makes this meme deeply offensive and discriminatory. The fourth meme (M4) relies on humor and sarcasm by contrasting how individuals perceive themselves with how others—specifically a socially revered neighbor—might see them. The image of a guarded, tough-looking man combined with the text's mocking tone creates a scenario that can erode self-esteem and foster social exclusion, depending on how it is interpreted by viewers.

These examples collectively demonstrate that the detection of cyberbullying in memes cannot rely on text or image alone. Visual cues often provide context for the target of the content, while the text shapes the sentiment and intent behind it. Humor, sarcasm, and cultural references further complicate the classification task, as implicit harm is often disguised in seemingly benign formats. Multimodal cyberbullying detection frameworks that integrate both text and image are therefore essential to accurately assess the harmfulness of such content, offering not only better detection but also more transparent explanations for why a meme is flagged as bullying.

The MultiBully dataset is a benchmark dataset comprising 5,798 Hinglish memes, with 3,075 labeled as bully and 2,723 as non-bully. This dataset comprises memes from open-source platforms like Twitter and Reddit. The dataset is designed to facilitate research on sentiment, emotion, and sarcasm-aware cyberbullying detection in multimodal, code-mixed settings. This dataset, with its diverse collection of multimodal memes, provides a valuable resource for training and testing cyberbullying detection models in the Hinglish language. For this study, we focus on the binary classification task (bully vs non-bully), using 80% of the data for training and 20% for testing, ensuring stratified splits to maintain class distribution. By leveraging the rich linguistic and visual information present in this dataset, the proposed approach aims to advance the state-of-the-art in cyberbullying detection and contribute to a safer online environment.

4.2 Evaluation metrics

We have used accuracy and F1 Score as the primary metrics to calculate and compare the performance of our framework against other baseline models. These metrics allow the researcher to compare multiple models and help in finding out the suitable one [33]. Further, these metrics are briefly explained below.

Accuracy It is the most straightforward way of evaluating the model as a ratio of the total number of correctly classified to the total number of data entries, as shown in the

below equation. It is simple to comprehend and interpret accuracy. It shows the percentage of cases that were successfully classified out of all of them. Because of its simplicity, it is a statistic that is easily accessed by researchers and practitioners alike, which facilitates the unambiguous communication of model performance. When classes are broadly balanced, or have about the same amount of samples in each class, accuracy works well. In these situations, it efficiently gauges the model's overall effectiveness without giving preference to any specific class.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Here,

TP True Positives (number of bully memes correctly classified as bully)

TN True Negatives (number of non-bully memes correctly classified as non-bully)

FP False Positives (number of non-bully memes incorrectly classified as bully)

FN False Negatives (number of bully memes incorrectly classified as non-bully)

F1 Score The F1 score, a widely used evaluation metric in binary classification tasks, is employed in this study to assess the performance of the proposed cyberbullying detection approach[34–36]. As a harmonic mean of precision and recall, the F1 score provides a balanced measure of the model's ability to correctly identify both bully and non-bully memes. This metric is particularly valuable in scenarios where class imbalance is present, as is often the case in real-world cyberbullying datasets. The F1 score ranges from 0 to 1, with a higher score indicating better overall performance. By focusing on the F1 score, this study prioritizes a model that achieves a good balance between precision and recall, ensuring that both types of misclassifications are minimized.

In this study, we calculate the macro F1 score, which is the average of the F1 scores for each class (bully and non-bully), giving equal weight to both classes regardless of their support. The macro F1 score is calculated as follows:

$$F1_{\text{macro}} = \frac{1}{2} \left(\frac{2 \cdot \text{precision}_{\text{bully}} \cdot \text{recall}_{\text{bully}}}{\text{precision}_{\text{bully}} + \text{recall}_{\text{bully}}} + \frac{2 \cdot \text{precision}_{\text{non-bully}} \cdot \text{recall}_{\text{non-bully}}}{\text{precision}_{\text{non-bully}} + \text{recall}_{\text{non-bully}}} \right)$$

4.3 Comparison models

This section discusses the baseline method and state of-the-art methods adopted to compare our proposed model performance. The proposed Teacher-Student Transfer learning model is compared to standard baselines of BERT+GRU+Fully Connected, Resnet+Fully Connected, BERT-Resnet+Fully Connected, CLIP+Fully Connected. The model parameters of the text and image encoders and decoders of baseline models are mentioned in Table 1.

BERT+GRU+Fully Connected (Text-Only):

This model uses BERT embeddings as input to a GRU (Gated Recurrent Unit) network, followed by a fully connected classification layer. It represents a typical sequence modeling approach for text-based classification tasks, leveraging pre-trained contextual embeddings but without any visual information.

ResNet+Fully Connected (Image-Only)

Table 1 Architectural parameter for text and image feature extraction

Domain	Model	Type	Output size
Text	Multilingual BERT	MBERT	50 x 768
	Multilingual BERT and BiGRU	BiGRU	50 x 512
	CLIP+Text Encoder	BERT	512
Image	ResNet and Dense	ResNet	7 x 7 x 2048
		Global Average Pool	2048
		Dense	512
	CLIP Image Encoder	Vision Transformer	512

It uses ResNet, a convolutional neural network pre-trained on image datasets, to extract image features followed by a fully connected layer for classification. This method provides a benchmark for how much information is captured from image features alone without considering textual cues.

BERT+ResNet+Fully Connected (Text-Image Concatenation)

It combines textual and visual features by concatenating the outputs from BERT and ResNet encoders before feeding them into a classification layer. This represents a basic fusion approach where modalities are integrated at the feature level without interaction modeling.

CLIP+Fully Connected (Text-Image Concatenation)

It employs CLIP (Contrastive Language-Image Pre-training) to extract embeddings for both text and image, which are concatenated and passed through a classification layer. CLIP's pre-training on aligned text-image data gives it better cross-modal representations than independent embeddings like BERT or ResNet.

BERT+ResNet+Fully Connected (Text-Image Inter-Attention)

It incorporates an inter-attention mechanism where textual and visual features attend to each other, enabling more nuanced cross-modal reasoning. This improves over concatenation by modeling interactions, allowing the model to focus on salient aspects of each modality relevant to classification.

CLIP+Fully Connected (Text-Image Inter-Attention)

It uses CLIP embeddings with an inter-attention mechanism to enable cross-modal interaction before classification. This combines the advantages of pre-trained vision-language models and attention mechanisms, representing one of the strongest baselines.

4.4 Hyperparameter selection

Our framework is built upon the mT5 encoder-decoder architecture, specifically the "mt5-base" version. Image features are extracted using the CLIP-ViT-B/32 vision Transformer, which remains frozen during training, as in previous work. To effectively combine textual and visual information, a single-head cross-attention mechanism is integrated into each layer of the mT5 encoder. The maximum input text length is set to 512 tokens. In the fusion process, text features serve as the query, while image features act as the key and value. These fusion modules are initialized randomly. The hidden state dimension (m) is set to 768, and the key dimension (mk) is set to 384. To ensure both robustness and reproducibility, we conducted a systematic grid search over key hyperparameters and selected configurations that yielded optimal performance while balancing computational efficiency as shown in Table 2.

Table 2 Hyperparameter selection and description

Hyperparameter	Selected value	Description
Number of Epochs	20	Allows sufficient training iterations while avoiding overfitting
Batch Size	32	Ensures stable gradient updates with efficient memory usage
Learning Rate	5×10^{-5}	Chosen empirically through grid search, balancing convergence speed and stability
Warmup Strategy	Linear	Gradually increases the learning rate during the initial phase to stabilize training
Warmup Steps	10% of total steps	Helps avoid abrupt gradient updates early in training
Maximum Input Length	512 tokens	Captures complete meme text and rationales while conforming to computational limits
Image Size	224×224 pixels	Standard dimension for vision transformers, preserving important features without excessive computational cost
Hidden Size (m)	768	Matches the mT5 base architecture's size, ensuring compatibility and optimal performance
Key Dimension (mk)	384	Optimized for efficient attention computation within cross-attention layers

Table 3 Performance comparison of various models for harmfulness detection on the MultiBully dataset. Best results are in bold

Model	Acc	F1
BERT+GRU+FC (Text Only)	60.86	60.33
RN+FC (Image Only)	62.51	62.14
BERT+ResNet+FC (Text-Image Concat)[37]	65.21	64.66
CLIP+FC (Text-Image Concat)[38]	66.71	65.89
BERT+ResNet+FC (Text-Image Inter-Attention)[39]	65.28	65.14
CLIP+FC (Text-Image Inter-Attention)[38]	66.91	66.28
Proposed framework	73.31	73.23

4.5 Result and discussion

In this section, we have discussed the performance of unimodal and other baseline model (as discussed in section 4.3) and compared their performance with the proposed model.

The experimental results underscore the efficacy of the proposed approach, achieving an accuracy of 73.31% and an F1 score of 73.23% as shown in Table 3. This performance is notable, as it surpasses several baseline models, including those that rely solely on text or image features, as well as those that do not employ the debate framework.

From Table 3, following finding are observed.

- Our framework outperforms Text-Only and Image-Only models by 12.45% and 10.8% in the Accuracy metric and 12.9% and 11.09% in the macro F1 score metric. The model's ability to outperform baselines that rely on unimodal features (either text or image alone) underscores the importance of considering the multimodal nature of memes. By incorporating both text and image information, the model can capture the interplay between these modalities, which is crucial for understanding the subtle cues and implicit meanings that often characterize harmful memes.
- Our framework also shows a major improvement over multimodal baseline approaches, including both Concatenation of Image and Text Features and Inter Attention Models, by at least 6.4% in Accuracy and 6.95% in F1 scores. This shows the effectiveness of our Teacher-Student Learning framework in capturing nuanced

interactions between textual and visual modalities, thereby enabling more accurate and context-aware detection of cyberbullying content

4.5.1 Impact of LLM rationales and noise analysis

While the LLM-generated rationales generally provide valuable context, we observed instances where they introduced noise. The rationales were most helpful in detecting *implicit* bullying, where the visual context (e.g., a specific gesture) altered the meaning of benign text. However, noise was introduced when the Teacher LLM hallucinated harmful intent in sarcastic but non-bully memes, or failed to grasp specific Indian culture references in the Hinglish text. In such cases, the rationale provided a “confidently wrong” context, which occasionally misled the Student mT5 model. Despite this, the preference selection mechanism (as mentioned in equation 1) largely mitigated this by filtering out the less plausible rationale before the student training phase.

To quantify these observations, we analyzed the model’s performance on specific subsets of the test data, as shown in Table 4. The results confirm that the LLM rationales provide the most significant benefit in the *Implicit Bully* category, improving accuracy by 16.5%. This supports our hypothesis that the “Teacher” effectively bridges the semantic gap where visual context alters the meaning of benign text. Conversely, the “Noise” factor is quantitatively visible in the *Sarcastic* category, where the proposed model saw a slight performance drop of 2.7% compared to the baseline. This indicates that the Teacher LLM occasionally over-analyzes sarcasm, generating “Bully” rationales for non-bullying jokes (hallucination), which misleads the student model. This quantitative trade-off, a massive gain in detecting hidden bullying at the cost of a minor increase in false positives on sarcasm, highlights the specific areas where the Teacher-Student transfer is most effective.

The significant performance of the proposed model highlights the importance of integrating both textual and visual information in cyberbullying detection tasks. Moreover, the results emphasize the effectiveness of the Teacher-Student framework, which enables the model to generate explanations from multiple perspectives, thereby capturing the nuanced and often implicit nature of harmful content in memes.

Furthermore, the superior performance of the proposed model compared to those that do not utilize the Teacher-Student framework highlights the benefits of generating explanations from multiple perspectives. The proposed framework allows the model to consider both harmless and harmful interpretations of a meme, leading to a more comprehensive understanding of its potential for harm. This approach not only improves the accuracy of the model but also enhances its explainability, as the generated explanations provide insights into the reasoning behind the model’s decisions.

Table 4 Comparative Performance Analysis (Accuracy %) when the LLM generated rationales provide the most help versus when they introduce noise across different meme categories

Category	Baseline (No Rationale)	Proposed (With Rationale)
Explicit Bully	71.50	74.20 (+ 2.70)
Implicit Bully	52.40	68.90 (+ 16.50 (Help))
Sarcastic (Non-Bully)	64.20	61.50 (- 2.70 (Noise))
Benign (Non-Bully)	75.10	76.80 (+ 1.70)

Table 5 Ablation study for cyberbullying detection

Model Variant	Accuracy (%)	F1 Score (%)
Proposed Model	73.31	73.23
Student-Only (No Teacher)	68.12 (+ 5.19)	67.89 (+ 5.34)
Concat (No Cross-Attention)	65.43 (+7.88)	64.95 (+ 8.28)

4.5.2 Ablative study

To better understand the contribution of each component in our proposed Teacher-Student Learning framework, we conducted an ablation study by systematically removing or modifying key components of the model. This analysis helps quantify the impact of each component on overall performance and provides insights into the relative importance of multimodal interactions, reasoning mechanisms, and input representations. We have experimented without the teacher guidance, relying solely on the student module to perform detection. In addition, to observe the significance of the cross attention mechanism, it is replaced with concatenation mechanism where textual and visual features are concatenated directly without any cross-attention mechanism.

From Table 5 following points are observed as:

- Removing the teacher module resulted in a drop of 5.19% in accuracy and 5.34% in F1 score, highlighting the importance of generating reasoning from multiple perspectives. This demonstrates that dialectical reasoning plays a crucial role in helping the student model better interpret ambiguous or context-dependent content.
- The simple concatenation approach resulted in a 7.88% drop in accuracy and 8.28% in F1 score, reinforcing that cross-attention mechanisms are vital for aligning and interpreting multimodal cues, especially in cases involving sarcasm, humor, or subtle imagery.

5 Conclusion and future work

The experimental results demonstrate the effectiveness of the proposed approach for multimodal cyberbullying detection in Hinglish memes, achieving an accuracy of 73.31% and an F1 score of 73.23%. This performance surpasses several baselines from the literature, including those that only consider text or image features, as well as those that do not utilize the classroom framework. The results highlight the importance of incorporating both text and image information, as well as the benefits of generating explanations from multiple perspectives. The superior performance of the proposed model compared to those that rely on unimodal features (either text or image alone) underscores the importance of considering the multimodal nature of memes. By incorporating both text and image information, the model can capture the interplay between these modalities, which is crucial for understanding the subtle cues and implicit meanings that often characterize harmful memes. Furthermore, the superior performance of the proposed model compared to those that do not utilize the classroom framework highlights the benefits of generating explanations from multiple perspectives. The classroom framework allows the model to consider both harmless and harmful interpretations of a meme, leading to a more comprehensive understanding of its potential for harm. This approach not only improves the accuracy of the model but also enhances its explainability, as the generated explanations provide insights into the reasoning behind the model's decisions.

It is also important to note the limitations regarding fine-grained semantic distinctions. While the Teacher-Student framework improves general detection, the Teacher

LLM is trained primarily on high-resource languages. Consequently, it occasionally struggles with the subtle, fine-grained semantic distinctions present in code-mixed Hinglish, such as local slang or community-specific satire. This suggests that while the "Teacher" can transfer reasoning capabilities, it cannot fully replace the need for domain-specific cultural knowledge in the underlying model.

In future work, we plan to explore several avenues to further enhance the performance and explainability of the proposed approach. First, we will investigate the use of more sophisticated fusion techniques to better integrate the text and image features. Second, we will experiment with different language models and strategies to improve the quality of the generated explanations. Third, we will explore the potential of incorporating additional modalities, such as audio and video, to further enhance the model's understanding of the nuances of harmful content. Finally, we will investigate the potential of the proposed approach for real-time cyberbullying detection in social media platforms, with the goal of developing a practical tool that can help to create a safer and more inclusive online environment.

Abbreviations

LLMs	Large language models
SVM	Support vector machines
mT5	Multilingual Text-to-Text transfer transformer
CNN	Convolutional neural network
RNN	Recurrent neural network
BERT	Bidirectional encoder representations from transformers
LM	Language model

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Author contributions

First author Pratik Joshi, implemented the code and drafted the initial manuscript. Coauthor Shikha Mundra contributed to the methodology and validation. Coauthor Ankit Mundra reviewed, edited, and approved the final version of the manuscript.

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Data availability

The dataset analysed during this study is publicly available at: <https://github.com/Jhprince/MultiBully?tab=readme-v-file>

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not Applicable.

Competing interests

The authors declare no conflict of interest.

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